

BIDISHA DASH

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EDUCATION

B.Sc (HONS.) AGRICULTURE [[View Degree Certificate](#)] 26/11/2021 – 31/05/2025

Siksha 'O' Anusandhan (Deemed to be University) | Odisha, India | Website: <https://www.soa.ac.in/>

- Grade: 8.69/10 (First Division with Distinction)
- Major Subjects: Agronomy, Horticulture, Soil Science, Plant Pathology & Nematology, Entomology, Plant Breeding & Genetics, Extension Education, Agricultural Engineering, Statistics & Computer Application

HIGHER SECONDARY EDUCATION

08/2019 – 08/2021

Uluberia High School | West Bengal, India

- Grade: 9.12/10

SECONDARY EDUCATION

2019

Uluberia Binapani Girls High School | West Bengal, India

- Grade: 8.92/10
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TEST SCORES

TOEFL iBT: Score: 97/120 (Test date: 08/08/2025) [[View Score Report](#)]

RESEARCH & PROJECTS

Sugarcane Eye Spot Disease Detection System | AWaDH-Annam AI, Indian Institute of Technology | 2026

- Developed a two-stage deep learning pipeline using EfficientNet-B3 for healthy-unhealthy binary classification (99.86% accuracy, 99.91% best validation F1) followed by YOLO11m segmentation for precise disease localization achieving 84.59% bounding box precision and mAP@50 of 86.19% on a 768×768 resolution validation set.
 - Built a novel field-collected dataset of eye spot (*Bipolaris sacchari*) addressing limited public availability of image data, combining farm-collected samples with curated online datasets, and performed manual polygon annotation of 16,506 unhealthy leaf images for instance segmentation training.
 - Integrated a recommendation engine and confidence analysis module delivering disease localization, severity ranking, environmental risk conditions, and management practice recommendations.
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EXPERIENCE

Internship | Agriculture Intern

13/07/2026-Present

ANNAM.AI | INDIAN INSTITUTE OF TECHNOLOGY ROPAR | Punjab, India |

Website: <https://www.annam.ai/>

- Contributing to web scrapping, field data collection, data cleaning, sorting, and annotation for active AI-for-Agriculture projects which includes disease/pest detection and a children's agri-education app.

AI 4 AgriTech Course + Internship Programme | Agritech Intern

18/05/2026 – 01/07/2026

INDIAN INSTITUTE OF TECHNOLOGY ROPAR | Punjab, India | Website: <https://ihub-awadh.in/ANNAM.AI> | **IIT Ropar-TIF (iHub-AWadh)** [[View Certificate](#)]

- Learned to build end-to-end AI pipelines for precision agriculture, spanning supervised/unsupervised ML, deep learning, computer vision, image segmentation (U-Net), object detection (YOLOv5), anomaly detection, and hyperparameter optimisation across agricultural datasets.
- Gained hands-on experience in crop yield prediction, disease-pest detection, and agricultural land zoning, working across SVM, ANN, YOLOv5, K-Means clustering, and SHAP/LIME for model interpretability.
- Developed an AI-powered Sugarcane eye spot disease detection system.

Skills- ML/DL Algorithms (Linear Regression, SVM, KNN, ANN, CNN, U-Net, YOLOv5, EfficientNetV2, Isolation Forest, K-Means, DBSCAN, Transformers), **Libraries & Frameworks** (Scikit-learn, TensorFlow, Keras, PyTorch, OpenCV, Albumentations, NumPy, Pandas, Matplotlib, Seaborn, SciPy), **Tools & Platforms** (Jupyter Notebook, Google Colab, Flask/FastAPI, TFLite), **Techniques** (Data Augmentation, Hyperparameter Tuning (Grid Search, Bayesian Optimisation), Image Segmentation, Object Detection, Anomaly Detection, Clustering)

Summer Bioinformatics Internship

01/03/2026-30/04/2026

Genome Guide | Virtual Internship [[View Certificate](#)]

- Completed practical training in using Python for bioinformatics, R programming for data analysis, Cytoscape for network biology applications, Autodock for molecular docking, SRA Toolkit for NGS data handling, and PyMOL for protein structure visualization.

MCKA Agro Consultancy Services | West Bengal, India | Technical Trainee

01/07/2025-31/12/2025

Website: <http://www.mckaagro.com/>

- Managed polyhouse cultivation of high-value crops (gerbera, bell peppers, cherry tomatoes).
- Handled pro tray seedling production of different vegetable and flower crops.
- Performed field scouting on 40 fields of farmers in order to diagnose the crop health and soil problems, and assisted senior consultants to develop adoptable solutions.
- Assisted senior consultants in conducting training programs.

Experiential Learning Programme

27/12/2024- 31/04/2025

Siksha 'O' Anusandhan (Deemed to be University) | Odisha, India

- EL-422: Seed Production Technology- Produced foundation maize seeds of variety CAL-1415 through proper field isolation, roguing, and clean harvesting practices.
- EL-434: Seed Quality Testing & Quality Enhancement- Conducted seed quality testing (germination, purity, vigor, and moisture) and tried different types of seed priming and other treatments to identify the most effective methods for enhancing performance.

Internship Program | Student Intern | Website: <https://rkvk.ac.in/>

02/09/2024 – 02/12/2024

Rathindra Krishi Vigyan Kendra, Visva-Bharati | West Bengal, India [[View Certificate](#)]

- Analyzed farmer-submitted soil samples for 12 key parameters (pH, EC, OC, macro-and micronutrients) for generating soil health cards (SHC).
- Produced organic green gram, prepared biostimulants, vermicompost, and azolla for improved soil fertility, microbial activity, yield, and disease-pest management.
- Supported in field research on effects of nano fertilizer doses on experimental rice plots.
- Surveyed over 80 farmers to promote pulse production among non-pulse growers and documented cultivation practices, challenges, and yield parameters from 34 pulse-growing farmers.

- Used Participatory Rural Appraisal tools to study the agro-socioeconomic condition of the village.
 - Conducted 2 training programs on Integrated Pest Management for vegetable crops and paddy straw mushroom with proper field-based demonstrations and distribution of instructional manuals.
 - Conducted an awareness program on government schemes and subsidies and educated farming families through a street rally, drama performance, and distribution of leaflets.
 - Made alternative farm plans for small and marginal farmers, to increase their income by integrating practices like mushroom cultivation and biofloc fish farming.
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TRAINING

MOOC on Artificial Intelligence in Agriculture | Online Certification Course 16/04/2026–15/05/2026
ICAR–National Academy of Agricultural Research Management (ICAR-NAARM), Hyderabad |

- Learned about Artificial Intelligence applications on agriculture, covering Machine Learning, Deep Learning, Generative AI, Agentic AI, IoT-based Precision Agriculture, Python programming, OpenCV, and AI literacy critical for ethical, technology-driven future.

End-to-End Google Earth Engine | Spatial Thoughts

- Implemented remote sensing workflows, time-series analysis and change detection, applied ML techniques on remote sensing datasets, and developed interactive geospatial apps using Google Earth Engine.

Python for Data Science and Machine Learning Bootcamp | Udemy | Online [[View Certificate](#)]

R Programming – R Language for Absolute Beginners | Udemy [[View Certification](#)]

SQL for Data Science | Great Learning [[View Certification](#)]

Practical Geospatial Geostatistical Modeling | Coursera, Case Western Reserve University | [[View Certification](#)]

PRESENTATIONS

Sugarcane Eye Spot Disease Detection System | AWaDH-Annam AI, Indian Institute of Technology Ropar | 2026

Nutrient, Processing and Value Addition in Millets | National Rice Research Institute | National Millet Expo | 2023

OUTREACH ACTIVITIES

- Swachh Bharat Mission Volunteer | 2022, 2023, & 2024
 - Save Soil Movement Participant | 2022
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